

16.7.1 Introduction

MCP History:

- For Coal & Lignite Mine:
- First notification by MoC on 27.8.2009
- Amendment 8.9.2009, 11.1.2012, 12.4.2012
- amendment 7.1.2013
- Govt. Guidelines dated **updated 18-06-2024**
- **For Other mines under purview of Mineral Conservation and Development Rules' 2017:**
- **Rule 22 to 27 of Mineral Conservation and Development Rules' 2017 for other than coal & Lignite mine.**
- 16 December 2019 guidelines for mining plan with MCP.
- 30 June 2022, manual for MCP by IBM.

16.7.1 Introduction

Coal/Lignite	Mineral (MCDR 2017)
1. Fixed Deposit in Escrow Account managed by CCO (An escrow account is a financial arrangement where a neutral third party holds assets or funds until certain conditions are met. 2. The third party, known as an escrow agent, releases the assets or funds only when the predetermined contractual obligations are fulfilled. 3. Escrow amount based on the wholesale price index (WPI) issued by Gol. An amount equal to the escrow amounts to be deposited each year throughout the mine life compounded @5% annually. 4. 15 lakhs/hectare for OCP 5. 2 lakhs/hectare for UGP 6. Approved by Coal Controller Officer under Ministry of Coal	1. Escrow Account 2. Minimum Amount to be deposited is total of 10 lakhs in all cases 3. 3 lakhs/hectare for mines where >175 employed in OCP or >75 employed in UG (such mines called category mine A) 4. 2 lakhs/hectare for other than above category mine A (called B) 5. Approved by IBM under Ministry of Mines

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Chapter-III Mine Closure Plans

- ✓ **3. Mine closure guidelines** aim to ensure responsible closure and rehabilitation of mining sites, minimizing environmental degradation, safeguarding public health, and promoting sustainable development by restoring the land to a condition suitable for future use of land or returning it to its near-natural state, while also holding project proponents accountable for their closure obligations.

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- ✓ **3.1. Mine Closure Plans:** Mine Closure Plans will have two components viz.
- ✓ (i) Progressive or Concurrent Mine Closure Plan, and
- ✓ (ii) Final Mine Closure Plan.
- ✓ Progressive Mine Closure Plan would include various land use activities to be done continuously and sequentially during the entire period of the mining operations, whereas the Final Mine Closure activities would start towards the end of mine life, and may continue even after the reserves are exhausted and/or mining is discontinued till the mining area is restored to an acceptable level.
- ✓ The Mine closure details of the Mining Plan should be oriented towards the restoration of land back to its original as far as practicable or further improved condition.
- ✓ However, for any mine abandoned/discontinued after the year 2009 without an approved mine closure plan or any mine having an approved mine closure plan abandoned after the partial extraction the project proponent is mandated to prepare and obtain approval for Temporary and Final Mine closure plans in line with the Guidelines of Management of mines discontinued/abandoned/closed before the year 2009 issued by Ministry of Coal on 28th October 2022, within one year of issuance of these guidelines.

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- ✓ **3.2. Just Transformation:** It refers to the equitable process of transitioning from traditional coal mining toward more sustainable and environmental friendly manners, ensuring that the environment is protected, the land is restored, and affected workers, communities, and regions are supported and empowered throughout the transformation. It involves recognizing and addressing the social, economic, and environmental challenges associated with mine closure activities. Every mine owner shall take all possible precautions for undertaking sustainable mining while accomplishing prospecting, mining and mine closure activities etc.
- ✓ **3.2.1 Restoration & Repurposing:** With aims to minimize the long-term ecological damage caused by mining and to ensure that the land can support various ecosystems, regarding the land, replanting native vegetation, restoring water bodies, and rehabilitating wildlife habitats, the Project proponent shall complete all activities related to technical, biological reclamation and repurposing (agriculture, pisciculture, eco-park, recreational, landscaping, waterbody conservation or creation as per Mission Amrit Sarovar, irrigation etc. wherever it is applicable) related to Just Transformation before issuing the Final Closure Certificate. The third party certifying the expenditure on these activities should ensure the same.

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✓ 3.2.2 People and communities:

- ✓ All social amenities and infrastructure created (like hospitals, schools, community centers etc.) should be handed over to the State Government.
- ✓ The office and other buildings of mine may also be utilized or developed as skill development centers or others.
- ✓ Efforts need to be systematically undertaken in each progressive closure period of 5 years to enhance skills, livelihoods, and living conditions of communities (PAPs) directly or indirectly dependent within the vicinity of the mine.
- ✓ The project proponent shall submit a 5 yearly report, outlining of the initiatives undertaken to skill development and sustain the livelihoods of the affected communities.
- ✓ It will be supplementary to the progressive closure claims submitted.

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- ✓ **3.2.3 Project proponents** shall strive to achieve the Nationally Determined contribution and set targets for the reduction of Scope II emissions in the Mining Plan by at least 25 percent of annual generation by renewable.
- ✓ Further, the Project Proponent shall promote electric vehicles (EVs) and gas-based vehicles in opencast coal mines to enhance sustainability, reduce environmental impact such as greenhouse gas emissions, and decrease dependency on diesel fuel.
- ✓ **3.3. Environmental management:** The project proponent shall ensure mitigation of the adverse impacts of mining activities on the environment. This entails scientific assessment, monitoring, and control measures to safeguard air, water, and soil quality, biodiversity conservation, and identification of sensitive areas and potential risks of land degradation. safe and sustainable diversion of the nala for uninterrupted natural water flow. The project proponent shall also make efforts to recharge groundwater and adhere to water quality standards as per statutes.

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- ✓ **3.4. Progressive mine closure status** shall be prepared **every 5 years from the beginning of the mining operations**. These plans would be **examined periodically in every five years** period and to be subjected to third-party monitoring by the agencies authorized by the Central Government (Refer Appendix-V) or any other institutes/ organizations/ agencies specified from time to time for the purpose.
- ✓ **3.5. Mine Closure Cost:** The total cost for carrying out such activities shall be estimated for assessment of mine closure cost of the mine involving progressive and final mine closure activities such as dismantling of structures/demolition and cleaning of sites, rehabilitation of mining machinery, plantation, physical/biological reclamation, landscaping, biological reclamation of left-out overburden dump, filling up of de-coaled void, for specified post environmental monitoring, supervision charges, power cost, protective and rehabilitation measures including their maintenance and monitoring, stowing for underground mines, miscellaneous charges barbed wire fencing as prescribed boundary wall all around the vulnerable area, etc. for the post-closure period.

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- ✓ **3.5.1. Escrow Amount Calculation:** A guideline for mine closure was issued based on the wholesale price index (WPI) as notified by the Government of India from time to time while preparing the Mining plan and Mine Closure Plan. These rates need modification based on the wholesale price index (WPI). The revised rate is Rupees Fifteen Lakh per hectare for opencast and Rupees Two Lakh per hectare for underground Mine. These rates will be considered as Base Rate to be escalated based on the WPIs as on 1st April of the corresponding financial year as declared from time to time by the Government of India.
- ✓ [Illustration: $\{(Rs\ 15\ lakhs \times WPIs\ as\ on\ 1st\ April) / (WPI\ as\ on\ April\ 2024\ i.e.\ 153.00)\} = Rupees\ in\ lakh,$ in case of Opencast project].
- ✓ The annual escrow amount is to be computed considering the total project area of the mine multiplied by the escalated rate (at the above-mentioned rates) and dividing the same by the balance production life of the mine in years. An amount equal to the escrow amounts to be deposited each year throughout the mine life compounded @5% annually.

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- ✓ **3.5.1. Escrow Amount Calculation:** The project proponent is required to deposit the said amount for each financial year by 30th September of the respective year, failing which an interest @0.5 percent/month will be charged on the amount to be deposited for that respective year.
- ✓ [For example, if the annual amount works out to ₹100, then (in the first year the amount to be deposited will be ₹100, in the second year Rs. $₹100 \times (1+5\%)^1$ in the third year $100 \times (1+5\%)^2$ and so on.]
- ✓ Further, in case of the mine, where an escrow account is already opened, the annual closure cost is to be computed considering the total project area at the above-mentioned rates minus the balance amount already deposited and dividing the same by the balance production life of the mine in years and annual cost as arrived should be compounded @5 % annually.
- ✓ In case of revision of Mine Closure cost on account of revision of Mining Plan, the mine closure cost as per the approved mining plan or the mine closure cost estimated as per the latest WPI, whichever is higher is to be considered.

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- ✓ **3.5.2. Financial Assurance:** The Mining Company/ Mine Owner as a part of Financial Assurance will open a **Fixed Deposit Escrow account, with the Coal Controller Organization** (on behalf of the Central Government) as the exclusive beneficiary prior to the commencement of any activities on the land/project area of the mine and shall submit the same to Coal Controller Organization (CCO) before the permission is given for opening the mine. The mining company shall cause the payment to be deposited at the rate computed as indicated at Para 3.5.1. The owner of the company may select the Schedule Bank where the Escrow account is to be opened and inform the same to the Coal Controller, CCO.
- ✓ **3.5.3. The Coal Controller shall get the WPI** (used for escalation of financial assurance amount at the time of formulation of Mining plan) updated as of 1st April of the corresponding financial year at the time of opening of Escrow account. The mine owner/company including all public/private sector companies shall deposit the yearly amount in a Schedule Bank in accordance with Para 3.5.1. Coal Controller shall also get the information, submitted under Para 3.4 & 4.8, verified and get the yearly financial assurance amount modified with respect to the latest WPI in accordance with Para 3.5.1.



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- ✓ **3.5.4. The money to be provided per hectare of total Project Area** for the purpose is to be deposited every year on commencement of any development activity on the land for the mine after opening a Fixed Deposit Escrow Account before obtaining mine opening permission from the Coal Controller. Mining company/owners including all Public Sector Undertakings shall deposit the yearly amount in a Scheduled Bank. If the Mine owners fail to deposit the required annual amount in accordance with Para 3.5.1, 3.5.2 & 3.5.3, the Government may withdraw the mining permission.

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✓ 3.5.5. Reimbursement of Escrow Amount:

- ✓ i) Up to 50% of the total deposited amount in the previous year excluding interest accrued in the ESCROW account may be released based on work done towards mine closure after every year. The release shall be based on
 - ✓ (a) Cost certification by certified auditor
 - ✓ (b) Certification of work done by the agencies authorized by the Central Government (Refer Appendix-V) or any other institutes/ organizations/ agencies specified from time to time for the purpose.
 - ✓ (c) High resolution georeferenced orthorectified multispectral satellite image or drone survey of the mine/blocks along with processed output submitted by the project proponent.
- ✓ The amount released should be equal to the expenditure incurred on the progressive mine closure in the past financial year or 50% of deposited amount, whichever is less.
- ✓ Project Proponent is required to submit the claim by the 30th of June and CCO will release the amount by 30th of September.

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✓ **3.5.5. Reimbursement of Escrow Amount:**

- ✓ ii) In line with periodic examination of the Closure Plan as per Para 3.4 & 3.7, up to 50 percent of the remaining amount including interest for the progressive closure period in the previous five years shall be released.
- ✓ However, in the year in which 5 yearly reimbursement is claimed, the yearly reimbursement will not be applicable i.e. the 1st year of every progressive closure period.
- ✓ The balance amount shall be released to mine owner/leaseholder at the end of the final Mine closure on compliance of all provisions thereof.
- ✓ An illustration in this regard is attached at Appendix-VI.
- ✓ The compliance report should be duly signed by the lessee and certifying that said closure of the mine complied all statutory rules, regulations, orders made by the Central or State Government, statutory organizations, court etc. and certified by the Coal Controller. Submission of claims & reimbursement shall be done through the dedicated portal. However, till the development of portal existing system of processing of claims will be continued.

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✓ **3.5.5. Reimbursement of Escrow Amount:**

- ✓ (iii) 90 percent of the balance amount at the end of the post mining closure period shall be released to the mine owner on compliance of all provisions of the Closure Plan duly signed by the mine owner/leasee to the effect that said closure of mine complied with all statutory rules, regulations, orders made by the Central or State Government, statutory organizations, court etc. and duly certified by the Coal Controller. This should also indicate the estimated extractable coal resources and coal actually mined out.
- ✓ (iv) A corpus from 10 percent of the balance deposited amount from final mine closure cost is to be created towards Just Transformation. This amount may be utilized by project proponent for socio transition after the closure of the mine in consultation with district administration, local authority and stakeholders.
- ✓ (v) In the case of fly ash dumping in abandoned mines where mine voids are separable as per Para 4.1 and a separate Mining Plan/Mine closure plans are prepared, only the apportioned amount for the required area for fly ash dumping such as the cost for Top Soil Management & Biological Reclamation, safety and environmental monitoring till the final mine closure and post closure is to be retained. The remaining amount may be released for rest of the area after the completion of post-mining closure activities.

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- ✓ **3.6. Final Mine Closure:** The details of the Mining Plan, covering the Final Mine Closure Plan envisaging the details of the updated cost estimates for various mine closure activities and the Escrow Account already set up, shall be submitted to the approving authority for approval at least five years before the intended final mine closure. Final Mine Closure Plan shall also contain a completion report containing operational history; past investigation and remedial efforts if any; mine working maps and drawings; mine water discharge and their uses, GPS coordinates for all salient features, safely accessible open portions of mine workings, dumps, area depillared, partially extracted and standing on pillars, any monitoring and maintenance satellite data and final landscape data etc.
- ✓ (a) Final Mine Closure would be considered to be completed only after acceptance of the third-party audit report by the Coal Controller on the compliance of all provisions of the Mine Closure Plan. Any Institute/ Organization/Agency as may be specified by the Government for this purpose may be engaged for Third Party audit to create a self-sustained ecosystem. Failure of restoration within the specified period may result in forfeiture of the Escrow Account created as per Para 3.5.1, and 3.5.2. The details of the Final Mine Closure Plan along with the details of the updated cost estimate for various mine closure activities and escrow account already set up shall be submitted at the time of approval of final mine closure plan.

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- ✓ **3.7. Time Scheduling for mine closure:** Post mining closure period is to be taken as 3 (three) years and post closure monitoring period will be for 2 (two) years after that. The Action plan for carrying out all mine closure operations (progressive and final mine closure) should be furnished in the form of a bar chart for the period of life of the mine plus post-closure period as specified in Appendix-VI but not limited to these activities. The project proponent is mandated to establish targets for closure activities according to the bar chart. Reimbursement of the ESCROW Amount during each progressive and final mine closure phase will be contingent upon the completion of closure work in accordance with the established targets. Project proponent is required to submit the compliance of established target annually to the coal controller.

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- ✓ **3.8. The funds so generated are towards the financial security** to cover the cost of closure in case the mine owner fails to complete the relevant closure activities. The prime responsibility of mine closure shall always lie with the mine owner, and in case these funds are found to be insufficient to cover the cost of final mine closure including the areas covered in Para 3.4, 3.5.1, 3.5.2, 3.5.3, & 3.6 above. The mine owner shall undertake to provide the additional fund equivalent to the gap in funding before five years of the end of coal production failing which it may be recovered by such other methods as the competent authority may deem fit in this regard.
- ✓ **3.9. Final Closure Certificate:** CCO will issue mine closure certificate to the effect that the protective, reclamation, rehabilitation work and work related to sustainability in accordance with the approved Mining plan covering final mine closure provisions/activities carried out by the mine owner. After successful implementation of the final mine closure activities and post closure monitoring, mine closure will be considered as completed.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

- ✓ **Rule No. 2. Applicability.-**
- ✓ These rules shall apply to all minerals, except –
- ✓ (i) petroleum and natural gas;
- ✓ (ii) coal, lignite and sand for stowing; and
- ✓ (iii) minor minerals:
- ✓ Provided that the State Government may by notification extend the application of any of the provisions of these rules, with such modification as may be deemed necessary and appropriate, to minor minerals:
- ✓ Provided further that the powers and responsibilities to be exercised by the Central Government or the Indian Bureau of Mines in respect of minerals other than minor minerals shall be exercisable by the State Government or State Directorate of Mining and Geology (by whatever named called) or any other authority subordinate to the State Government as may be notified by the State Government while extending the application of these rules to minor minerals.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

- ✓ **Rule No. 55 (2) (a)**
- ✓ **(a) category 'A' mines means–**
- ✓ (i) such fully mechanised mines where the work is being carried out by deployment of heavy mining machinery for deep hole drilling, excavation, loading and transport; or
- ✓ (ii) such mines where the number of average employment exceeds one hundred and fifty in all or seventy-five workings below ground, or mines where any of the mining operations like deep hole drilling, excavation, loading and transport is carried out with the help of heavy machinery;
- ✓ **(b) category 'B' mines means mines other than category 'A' mines:**
- ✓ Provided that if any doubt arises as to whether any mine is a category 'A' mine, it shall be referred to the Controller General for decision.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

✓ **Rule No. 10. Competent Authority.–**

- ✓ (1) The Controller General or an officer of Indian Bureau of Mines authorised in writing by the Controller General shall be the competent authority to take decisions with respect to a mining plan approved by any officer of the Indian Bureau of Mines, pursuant to clause (b) of sub-section (2) of section 5.
- ✓ (2) The State Government shall be the competent authority to approve a mining plan required for grant of a mining lease, prepared and certified in accordance with the system established by the State Government pursuant to the proviso to clause (b) of sub-section (2) of section 5 and approved by the Central Government: Provided that the Controller General or an officer of Indian Bureau of Mines authorised in writing by the Controller General shall be the competent authority to take decisions with respect to monitoring and implementation of such mining plans approved by the State Government.
- ✓ (3) The Director, Atomic Minerals Directorate for Exploration and Research or an officer of the Atomic Minerals Directorate for Exploration and Research authorised in writing by the Director, Atomic Minerals Directorate for Exploration and Research shall be the competent authority to take decisions with respect to a mining plan pursuant to clause (b) of sub-section (2) of section 5, in respect of minerals specified in Part B of the First Schedule to the Act, where the grade of such atomic minerals is equal to or above the threshold value limits declared under Schedule-A of the Atomic Minerals Concession Rules, 2016.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

✓ **Rule No. 22. Mine Closure Plan.-**

✓ (1) Every mine shall have mine closure plans, which shall be of two types; namely:—

✓ **(i) a progressive mine closure plan; and**

✓ **(ii) a final mine closure plan.**

✓ (2) Every holder of a mining lease shall take steps to prepare mine closure plans as per the guidelines and format given by the Indian Bureau of Mines from time to time: Provided that where the State Government has set up a system for preparation, certification and monitoring of mining plan pursuant to the proviso to clause (b) of sub-section (2) of section 5, such guidelines and formats shall be prescribed by the State Government: Provided further that in respect of minerals specified in Part B of the First Schedule to the Act, where the grade of such atomic minerals is equal to or above the threshold value limits declared under Schedule-A of the Atomic Minerals Concession Rules, 2016, such guidelines and formats shall be prepared in the format as may be specified by the Director, Atomic Minerals Directorate for Exploration and Research.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018): Rule No. 22 to 27

- ✓ **Rule No. 23. Submission of progressive mine closure plan.-**The holder of a mining lease shall submit to the competent authority a progressive mine closure plan as a component of the mining plan, at the time of submission, modification and review of the mining plan.
- ✓ **Rule No. 24. Submission of final mine closure plan.-**
- ✓ (1) The holder of a mining lease shall submit a final mine closure plan to the competent authority for approval two years prior to the proposed closure of the mine.
- ✓ (2) The competent authority shall convey his approval or refusal of the final mine closure plan within ninety days of the date of its receipt to the holder of the mining lease.
- ✓ **Rule No. 25. The modification of mine closure plan.-**
- ✓ (1) The holder of a mining lease desirous of seeking modifications in the approved mine closure plan, shall submit to the competent authority for approval setting forth the intended modifications and explaining the reasons for such modifications.
- ✓ (2) The competent authority may approve the modifications as submitted under clause (1), or approve with such alterations as he may consider expedient.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

- ✓ **Rule No. 26.** Responsibility of holder of a mining lease.- (1) The holder of a mining lease shall have the responsibility to ensure that the protective measures including reclamation and rehabilitation works have been carried out in accordance with the approved mine closure plan or with such modifications as approved by the competent authority. (2) The holder of mining lease shall submit to the competent authority a yearly report as per the format specified by the Indian Bureau of Mines, before 1st day of July every year setting forth the extent of protective and rehabilitative works carried out as envisaged in the approved mine closure plan, and if there is any deviation, reasons thereof: Provided that where the State Government has set up a system for preparation, certification and monitoring of mining plan pursuant to the proviso to clause (b) of sub-section (2) of section 5, such format shall be prescribed by the State Government: Provided further that in respect of minerals specified in Part B of the First Schedule to the Act where the grade of such atomic minerals is equal to or above the threshold value limits declared under Schedule-A of the Atomic Minerals Concession Rules, 2016, such format shall be prescribed by the Director, Atomic Minerals Directorate for Exploration and Research.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

✓ **Rule No. 27. Financial assurance.–**

- ✓ (1) A financial assurance shall be furnished by the holder of the mining lease, for due and proper implementation of the progressive mine closure 14 plan contained in the mining plan or the final mine closure plan, as the case may be, which shall be an amount of three lakh rupees for Category 'A' mines and two lakh rupees for Category 'B' mines, per hectare of the mining lease area put to use for mining and allied activities: Provided that the minimum amount of financial assurance to be furnished under sub-rule (1), shall be ten lakh rupees for Category 'A' mines and five lakh rupees for Category 'B' mines: Provided further that the provisions of sub-rule (1) shall not be applicable for a mining lease granted through the auction or the mining lease granted under the provisions of clause (b) or clause (c) of sub-section (2) of section 10A, wherein the Mine Development and Production Agreement has been signed between the lessee and the State Government.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

✓ **Rule No. 27. Financial assurance.–**

- ✓ (2) Where financial assurance is required to be furnished by the holder of the mining lease under sub-rule (1), such amount of financial assurance shall be submitted to the authorised officer, as the case may be, in the form of a bank guarantee in the format specified by the Indian Bureau of Mines: Provided that where financial assurance has already been furnished before the commencement of these rules, an amount equal to the difference between the financial assurance due as on the date of notification of these rules and the financial assurance already furnished, shall be furnished to the authorised officer, as the case may be, within a period of ninety days from the date of notification of these rules: Provided further that the holder of a mining lease shall be required to enhance the amount of financial assurance with the increase in the area of mining and allied activities: Provided also that where a leaseholder undertakes reclamation and rehabilitation measures as part of the progressive closure of mine, the amount so spent shall be reckoned as sum of the financial assurance already spent by the leaseholder and the total amount of financial assurance, to be furnished by the lessee, shall be reduced to that extent.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

✓ **Rule No. 27. Financial assurance.–**

- ✓ (3) Release of financial assurance shall be effective upon the notice given by the lessee for the satisfactory compliance of the provisions contained in the final mine closure plan and certified by the authorised officer, as the case may be.
- ✓ (4) If the authorised officer, as the case may be, has reasonable grounds for believing that the protective, reclamation and rehabilitation measures as envisaged in the progressive mine closure plan contained in the approved mining plan or the final mine closure plan, in respect of which financial assurance was furnished, has not been or will not be carried out in accordance with the mining plan or the final mine closure plan, either fully or partially, the authorised officer, as the case may be, shall give the lessee a written notice of his intention to issue the orders for forfeiting the sum assured atleast thirty days prior to the date of the order to be issued.

Mineral Conservation and Development Rules 2017 (amended up to 27th March 2018)

Rule No. 22 to 27

✓ **Rule No. 27. Financial assurance.–**

- ✓ (5) Within thirty days of the receipt of notice referred to in sub-rule (4), if no satisfactory reply has been received in writing from the lessee, the authorised officer, as the case may be, shall pass an order for forfeiting the surety amount and a copy of such order shall be endorsed to the concerned State Government.
- ✓ (6) Upon the issuance of order by the authorised officer, as the case may be, the concerned State Government may realize the bank guarantee provided or obtained as financial assurance for the purpose of performance of protective, reclamation, rehabilitation measures and shall carry out those measures, or appoint an agent to do so.

16.7.1 Introduction

- ✓ As well as the various phases such as exploration, development and exploitation, mine closure is equally important.
- ✓ Regulations have been enforced in almost all countries to comply with this aspect.
- ✓ Mine owners are required to commit the execution of this phase during mine planning and while submitting the Environment Impact Statement (EIS) to the Government.
- ✓ This means it has got financial implications, and this is an extra cost, which was almost absent in the past, say prior to 1990.
- ✓ The predicted adverse impacts must be reported in EIS and appropriate means for their mitigation must be proposed.
- ✓ A systematic and methodical approach is favored so that likely impacts are clearly identified and there is no doubt that all potential impacts have been considered.
- ✓ This involves preparation of a detailed mine closure programme that addresses all the issues associated with closure and rehabilitation.

16.7.1 Introduction

- ✓ The objective of the closure plan is to assure the controlling authorities that it will be successful and to release the mine owners and operators from their obligations so that the site can be disposed of in an appropriate manner.

16.7.2 Phases – Mine closure

- ✓ Mine closure programmes comprise three phases:
 - ✓ ● Closure Planning, immediately prior to cession of operations;
 - ✓ ● Active Care, during which the mine is decommissioned and the whole site rehabilitated; and
 - ✓ ● Passive Care, a period of monitoring to show the rehabilitation has been successful and that the site can be disposed of.
- ✓ A bar chart shown in figure 16.46, illustrates these aspects.
- ✓ It is very unusual nowadays for a mining and mineral operations to get planning approval without the provision of some means of financial surety to indemnify the authorities against closure and rehabilitation costs. Different mechanisms are available, and the choice will depend on the preference and circumstance of the owner and the financial advice that he receives.

PLANNING FOR MINE CLOSURE

16.7.2 Phases – Mine closure

Figure 16.46 Schedule of mine closure operations (www.smenet.org).

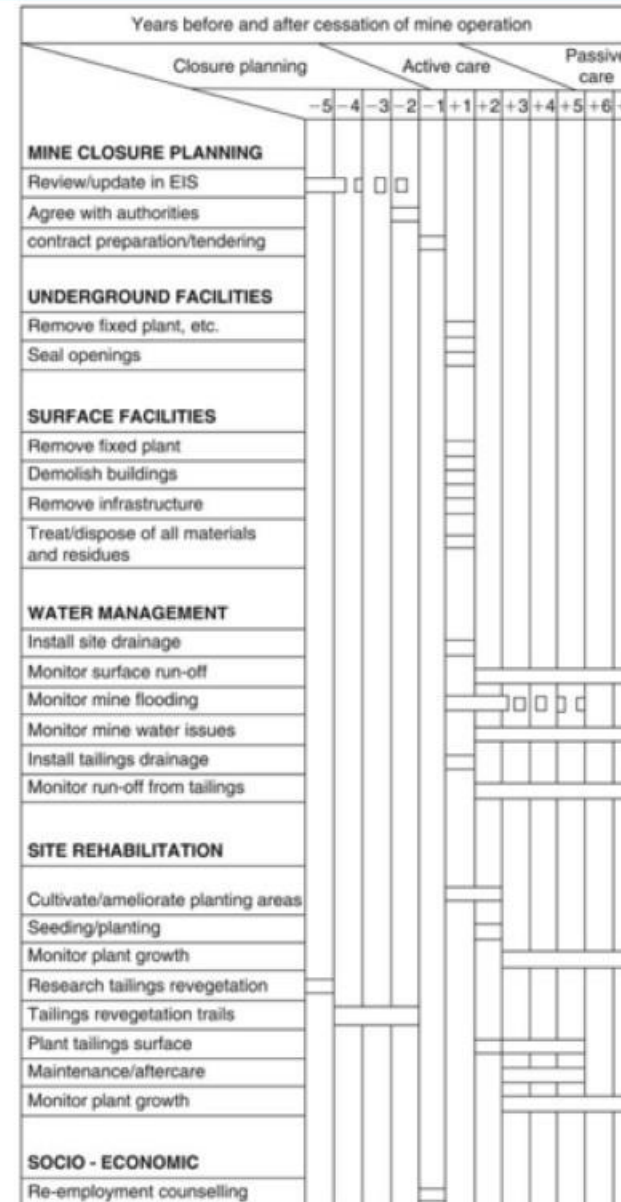


Figure 16.46 Schedule of mine closure operations (www.smenet.org).

PLANNING FOR MINE CLOSURE

16.7.3 The Integrated Mine Closure Planning Guidelines

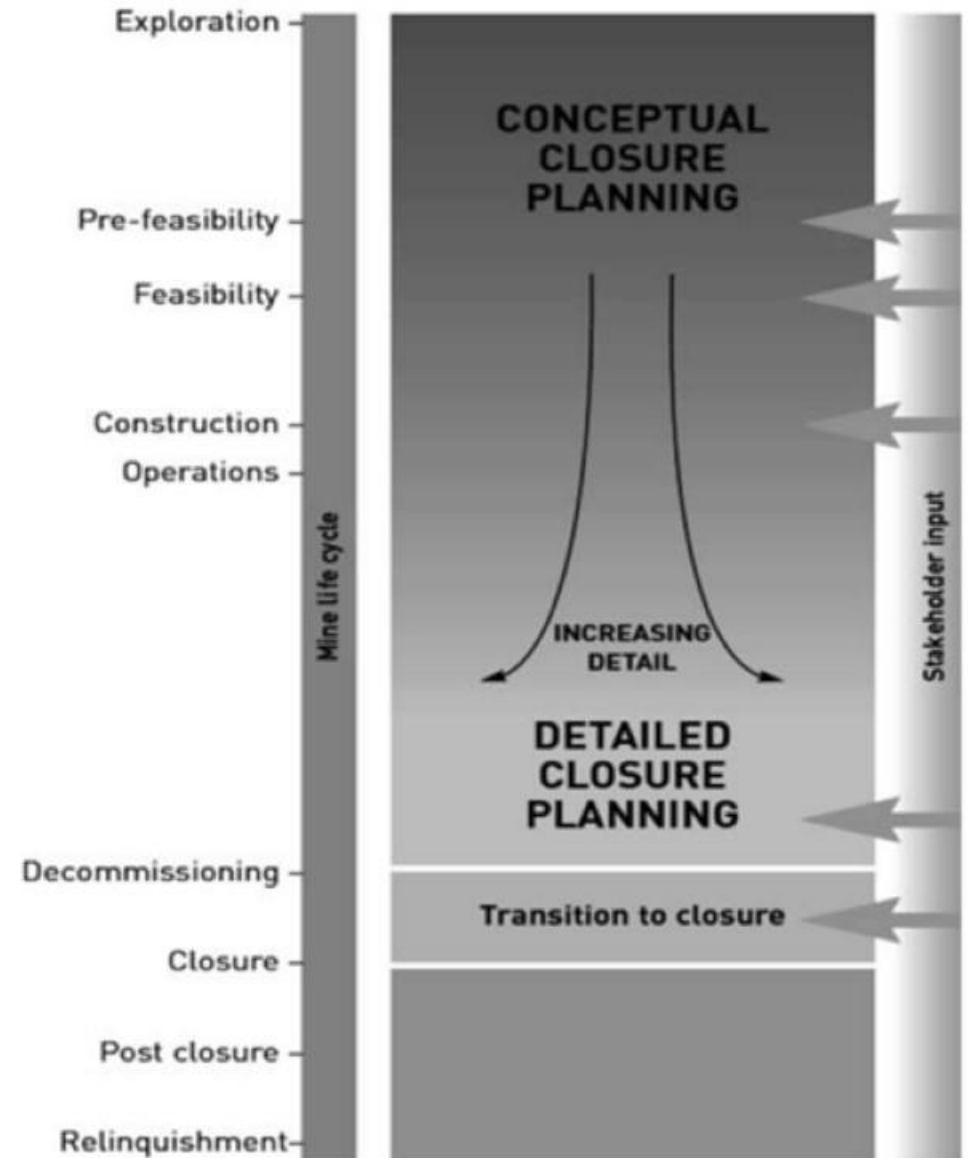


Figure 16.47 Closure planning (Permission: ICMM).

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

- ✓ The Integrated Mine Closure Planning Guidelines (Toolkits) were developed by ICMM to help site consideration of closure aspects in a holistic manner.
- ✓ The following text, with kind permission from ICCM, is presented herewith.
- ✓ This concept, and the supporting material in the report, apply equally well to both large and to small companies.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ **1. Expected outcome:** Positive outcomes of effective closure planning should mean that:
- ✓ ● engagement with affected and interested parties will be more consistent and transparent;
- ✓ ● communities will participate in planning and implementing actions that underpin successful closure;
- ✓ ● closure decisions will be better supported by stakeholders;
- ✓ ● planning for closure will become easier to manage;
- ✓ ● the accuracy of closure cost estimates will be improved;
- ✓ ● the risk of regulatory non-compliance will be minimized;
- ✓ ● potential problems will be identified in a timely manner;
- ✓ ● there is more likely to be adequate funding for closure;
- ✓ ● potential liabilities will be progressively reduced; and
- ✓ ● opportunities for lasting benefits will be recognized and planned for adequately.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ **2. Periodic checks and review:**
- ✓ Successful closure depends on setting, continually reviewing and validating and finally meeting closure goals that align with company and stakeholder requirements. There should be minimal residual risk to the company, and the community should realize benefits that will continue to exist without further input from the company.
- ✓ **3. Number of phases within the lifecycle:** The life cycle of the operation has been characterized as having eight phases, as shown in figures 16.48 & 16.49: exploration, pre-feasibility, feasibility (which includes planning and design), construction, operation, decommissioning, closure and post closure (which may include relinquishment of tenure and liability).

PLANNING FOR MINE CLOSURE

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

✓ 3. Number of phases within the lifecycle:

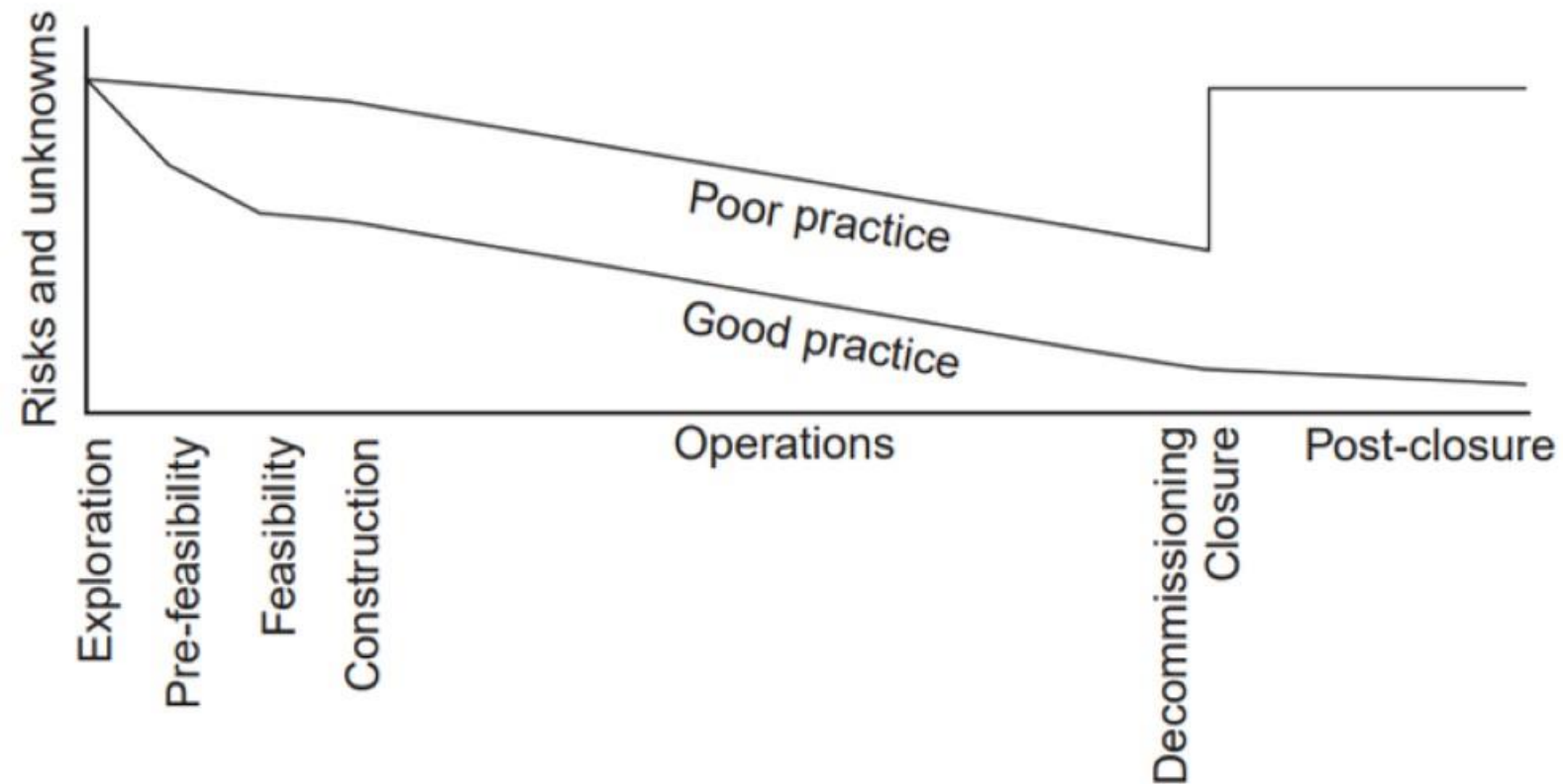


Figure 16.48 Reduction of risks and unknowns (Permission: ICMM).

PLANNING FOR MINE CLOSURE

16.7.3.1 Salient features (parameters to be considered) for closure planning

3. Number of phases within the lifecycle:

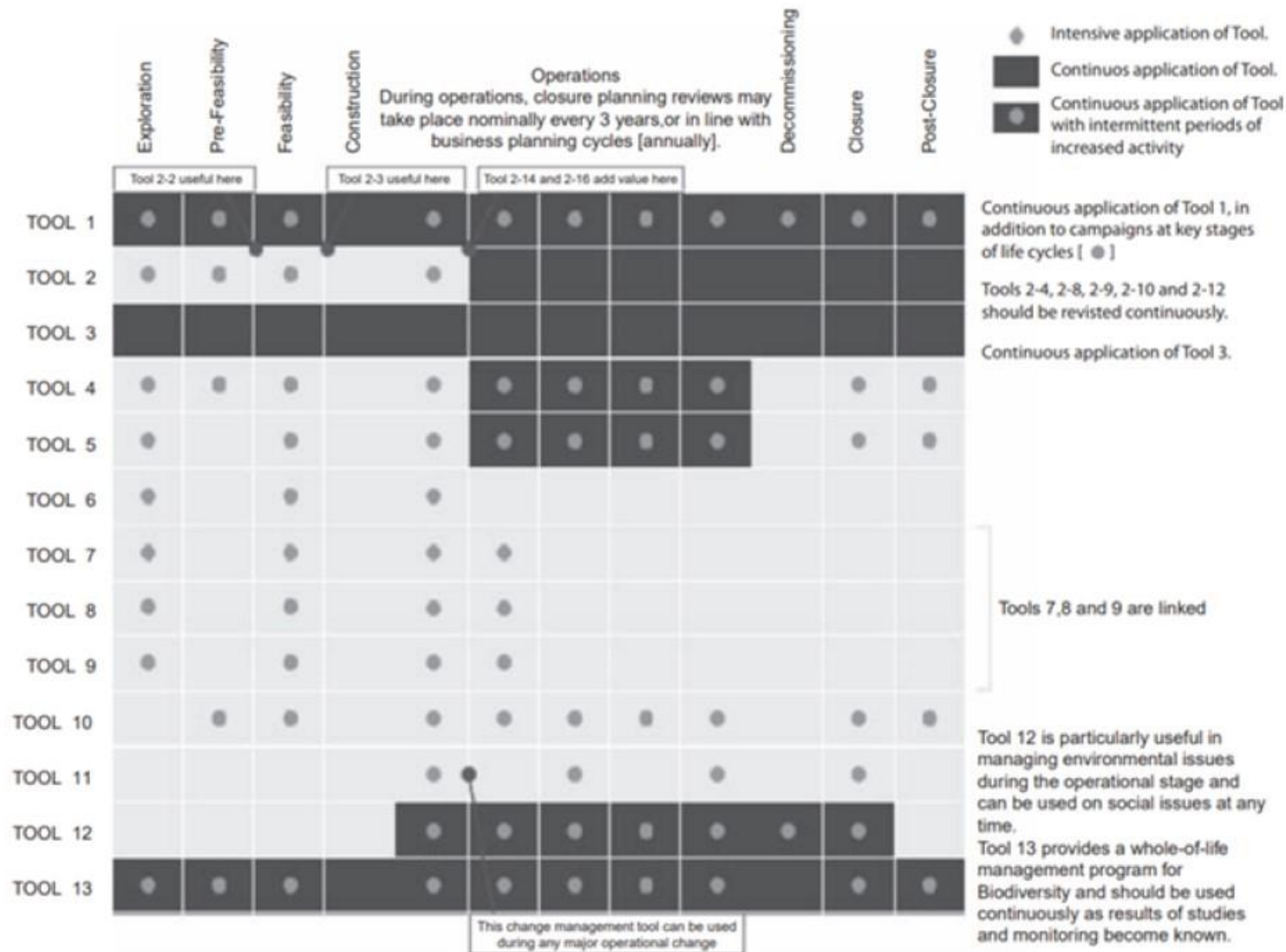


Figure 16.49 Integrated Mine Closure Planning Guidelines (Toolkits) as applicable throughout the life of an operation (Permission: ICMM).

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ **4. Basic steps to be considered:** There are three basic steps to developing an effective closure plan. If a mine makes closure planning part of its operational philosophy, these steps should blend into each other over time rather than being distinctive stages.
- ✓ I. Development of a conceptual closure plan: This plan is developed and used during exploration, pre-feasibility, feasibility/design and construction to guide the direction of activities. Its active life may be three to five years.
- ✓ II. Development and implementation of a detailed closure plan: This plan is used continuously during operations, and has an active life that could range from 5 to 30 years or more, during which time it is updated. A detailed plan will include milestones, detailed methodologies of achieving these, monitoring and validation processes.
- ✓ III. The final step is decommissioning and post closure plan: Its active life may be as little as a year or two, although depending on post closure responsibilities, it may extend many years past that time.

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16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ **5. Participants in effective closure planning:** Effective closure planning involves bringing together the views, concerns, aspirations, efforts and knowledge of various internal and external stakeholders to achieve outcomes that are beneficial to the operating company and the community that hosts it. For a company, this involves:
 - ✓ ● Incorporating closure planning into the early stages of project development (nominally pre-feasibility and feasibility) and operations;
 - ✓ ● Collating the goals and views of various stakeholders (project owner (internal stakeholder***), external stakeholders such as: local community, government, and non-governmental organizations (NGOs)) at the early stage of project development and operations to inform closure and post closure goals;
 - ✓ ● Acting to meet the goals by working with the relevant stakeholders within and outside the project owner's organization;
 - ✓ ● Using the concepts of risk and opportunity to both minimize liability and maximize benefits to all relevant parties; and

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

✓ 5. Participants in effective closure planning:

- ✓ ● Using multidisciplinary expertise and multi-stakeholder processes to ensure that mitigation of risk in one area does not increase risks in another.

***Internal stakeholders

- ✓ The resources of an organization can have varying inputs and responsibilities, at different times, for the three stages of effective closure planning (conceptual closure planning, detailed closure planning and decommissioning/post closure planning).
- ✓ The need for integration between the various internal stakeholders at any one stage and between the stages is fundamental in ensuring that a mine operation is designed with closure in mind.
- ✓ Table 16.12 shows an example of how important internal stakeholders can have a key input to the closure planning process throughout the operation's life cycle.
- ✓ Note that some organizations may not have all of these internal divisions or teams to provide the requisite support, but much of the knowledge and expertise is usually drawn upon during the various stages of the development and operation of the facility.

5. Participants in effective closure planning:

***Internal stakeholders

Table 16.12 Key inputs by the internal stakeholders to the closure planning process throughout the operation's lifecycle (Permission ICMM).

Typical organizational involvement (guide only)								
Stage of operation life cycle	Time interval	Exploration team	Corporate team	Feasibility team	Construction closure team	Operational team	Decommissioning/ closure team	Form of closure plan actively developed and used
Exploration	2–5 yrs	Core	Support	N/A	N/A	N/A	N/A	Conceptual closure plan
Pre-feasibility	1–5 yrs	Handover	Support	Core	N/A	N/A	N/A	Conceptual closure plan (developed further)
Feasibility	1–5 yrs	Handover	Core/Support	Core	Advisory	Advisory	N/A	Conceptual closure plan (developed further)
Construction	1–2 yrs	N/A	Advisory	Support	Core	Advisory	N/A	Conceptual closure plan (developed further)
Operation	5–30 yrs	N/A	Support	Advisory	Advisory (initial)	Core	N/A	Detailed closure plan
Decommissioning	0–1 yrs	N/A	Support	N/A	N/A	Core	Core	Decommissioning & post closure plan
Post-closure	5+ + yrs	N/A	Support	N/A	N/A	N/A	Core/Support	Decommissioning & post closure plan

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16.7.3.1 Salient features (parameters to be considered) for closure planning

5. Participants in effective closure planning:

***Internal stakeholders

- ✓ Note too that these responsibilities may overlap to some extent, depending on the organizational structure of a company.
- ✓ Table 16.12: Influence of project teams in delivering integrated closure planning. Definition of terms:
- ✓
 - A core activity is one in which the involved party is required to drive the process and may hold accountability for the success of that process.
 - A support activity is one in which the involved party takes an active role, providing cross-functional expertise or management input.
 - An advisory activity is one in which the involved party provides contextual information that is of value to the planning.
 - A handover activity is a transitional role to ensure that continuity of the closure planning process is maintained.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ **6. This concept of continuous closure planning** is not the same as concurrent rehabilitation. The former is a process that extends throughout the life of the mine. The latter is only one part of the closure planning process which is usually accounted to operations.
- ✓ **7. Right time to start the closure planning:** Realizing closure goals requires a progressive reduction of risks and unknowns, as shown in figure 16.48. The earlier that risks and unknowns are reduced, the greater the potential for meeting specific objectives. This is one reason that planning for closure should begin at the earliest opportunity.
- ✓ **8. Application of the Integrated Mine Closure Planning Guidelines (Toolkits):** These can be applied throughout the life of an operation, as shown in figure 16.49.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.1 Salient features (parameters to be considered) for closure planning

- ✓ These 13 Guidelines, as outlined below, provide the practitioner with practical work processes, examples and contexts within which to apply closure planning discipline:
- ✓ ● Guideline (Toolkit) 1: Stakeholder Engagement
- ✓ ● Guideline (Toolkit) 2: Community Development
- ✓ ● Guideline (Toolkit) 3: Company/Community Interactions to Support Integrated Closure Planning
- ✓ ● Guideline (Toolkit) 4: Risk/Opportunity Assessment and Management
- ✓ ● Guideline (Toolkit) 5: Knowledge Platform Mapping
- ✓ ● Guideline (Toolkit) 6: Typical Headings for Contextual Information in a Conceptual Closure Plan
- ✓ ● Guideline (Toolkit) 7: Goal Setting
- ✓ ● Guideline (Toolkit) 8: Brainstorming Support Table for Social Goal Setting
- ✓ ● Guideline (Toolkit) 9: Brainstorming Support Table for Environmental Goal Setting
- ✓ ● Guideline (Toolkit) 10: Cost Risk Assessment for Closure
- ✓ ● Guideline (Toolkit) 11: Change Management Worksheet
- ✓ ● Guideline (Toolkit) 12: The Domain Model
- ✓ ● Guideline (Toolkit) 13: Biodiversity Management

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 1: Stakeholder engagement** The application of stakeholder engagement to complex issues requires bringing experience and expertise to bear on the issues. However, the scoping and planning of stakeholder engagement can be guided by informed views on the various project development and operational processes. **Guideline (Toolkit) 2: Community development** The ICMM Community Development Guidelines (2006), which address one of the primary vehicles of lasting benefits, namely the development of host communities. (Individual Guidelines are highlighted here within Guideline 2 with the numbering system cross-referenced to the source document.)
- ✓ **Guideline (Toolkit) 2-1 (Stakeholder identification Guideline)** allows the key stakeholders to be identified. This Guideline should be used at all stages of conceptual and detailed closure planning to initially identify stakeholders and then to ensure that the stakeholder list is kept current. **Guideline (Toolkit) 2-2 (Social baseline study Guideline)** lists various aspects of social baseline studies that should be covered. The social baseline study contributes vital information relating to the sustainability of closure goals, particularly as they relate to the social and socio-economic conditions of the local area and the region. Note that for facilities with a long life (20 years or more) the baseline information collected may become outdated, and consideration should be given to updating specific social and socio-economic data sets to better inform the closure planner of changes and trends that may have occurred.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 2-3 (Social impact and opportunities assessment Guideline)** is particularly potent in developing closure planning goals and milestones. Positive and negative impacts are assessed, and this provides the closure planner with information that assists in addressing sustainable improvements in communities. Note the value of assessing positive social and socio-economic impacts and focusing the closure plan on ways to enhance or sustain them.
- ✓ **Guideline (Toolkit) 2-4 (Competencies assessment)** is useful in planning for the delivery of community development programs and assists in identifying the competencies required within the community to continue the functions within these programs after the facility has closed and its support terminated. The competencies assessment can also be used in addressing skilled, semi-skilled and unskilled worker remobilization at the end of a facility's lifespan.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 2-8 (Problem census)** provides a rapid means of assessing the priorities within communities and is suited to early engagement at the exploration phase of mining, for example, as it allows the organization to focus on immediate issues. It is also a useful Guideline to use between the formal reviews of the detailed closure plan, as it provides an efficient way of confirming that the closure planning activities still meet community requirements or of indicating where deviations might begin to occur. The Problem Census is particularly effective when used prior to the commencement of capital or labor intensive programs that represent a significant cost to the company, as it may forestall wasted effort or wasted capital.
- ✓ **Guideline (Toolkit) 2-9 (Opportunity ranking)** is useful in establishing priorities for community development programs, which allows better benefit/cost outcomes to be targeted early in the development of a mine. As with most social and community development agendas, priorities within communities can alter, and opportunities that previously represented high benefit/cost outcomes may lose their ranking.
- ✓ **Guideline (Toolkit) 2-10 (Stakeholder analysis)** provides a useful set of questions to facilitate a strong foundation in the stakeholder engagement process. For ongoing processes such as closure planning, which need continuous engagement, revision and updates, the stakeholder analysis Guideline allows the closure planner to make informed decisions on which goals of the closure plan are likely to be stable, which are subject to external influences and the extent to which the external influences might dictate the work program and milestones set.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 2-12 (Partnership assessment)** allows the facility to more critically evaluate the prime vehicle for sustainable development in communities, the use of partnerships. The assessment Guideline is equally adaptable to community partners, industry partners, government and NGOs. Many community development programs falter because of inadequate partnering processes, requiring the company to take a more proactive stance and expend greater resources in achieving closure goals. The partnership assessment Guideline should be used to re-evaluate partnerships during review processes as well as at the outset of partnership development.
- ✓ **Guideline (Toolkit) 2-15 (Logical framework)** should be used to develop the interim goals and milestones that underpin the overall goals and outcomes. Used in conjunction with Guideline 2-14 (Overview of monitoring and evaluation Guidelines) and Guideline 2-16 (Indicator development), it allows the systematic progression to postclosure outcomes and allows the user to define key performance indicators. The logical framework Guideline allows the execution strategy of closure planning to be managed. Because many of the closure planning activities are undertaken during operations, the logical framework Guideline suits a number of line management functions and can be used to complement ISO14001 programs for environmental management in operations when coupled with monitoring and evaluation Guidelines and indicator development.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 3 Company/community interactions to support integrated closure planning** The relationship between a company and its neighbouring communities develops over time. In the mining industry, the company undergoes a series of transitions prior to its operational phase, and this requires the relationship to be built within a dynamic environment. Building trust and mutual respect, particularly during the usually rapid series of transitions between exploration and operation, is a challenging issue. It is also one of the most important activities to pursue. It is in this series of transitions that the tone of relationships for the first years of an operation's life is set. Table 16.13 shows a suggested minimum platform for engagement and the actions that a company might take at the different phases of the operation's life to build trust and mutual respect in closure planning.
- ✓ **Guideline (Toolkit) 4 Risk/opportunity assessment and management** Likelihood scales Note [as shown in tables 16.14(a); (b) & (c)] that these are examples and suggestions; they can be redefined to suit the circumstances and risk sensitivity of any organization.

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16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.13 Proposed minimum platform for engagement and the actions during different phases of the operation's life to build trust and mutual respect in closure planning (Permission: ICMM).

Phase	Characteristics	Project owner actions that support integrated closure planning	Exploration team	Corporate team	Pre-feasibility team	Feasibility team	Operational team	Decommissioning team
Feasibility	• Short time frame • Little knowledge of continuity • First impressions • Rising community expectations	- Engage with communities - Understand expectations - Build rapport - Provide direction and standards - Provide background information	Y					
	• Limited time • Intensive studies • Scoping activity • Ability to discuss continuity • 'First strategies' • Rising community expectations	- Headframe - Engage with communities and stakeholders - Develop target closure outcome - Review objectives - Research and data - Build rapport - Estimate closure costs - Multidisciplinary decision-making - Provide direction and standards - Provide background information	Y		Y			
Feasibility	• Extensive studies • Extensive engagement • Discuss needs and priorities • Discuss mine options • Discuss development scenarios	- Headframe - Engage with communities and stakeholders - Fill knowledge gaps - Refine target closure outcome - Refine objectives - Begin achieving objectives	Y		Y			
Phase	Characteristics	Project owner actions that support integrated closure planning	Exploration team	Corporate team	Pre-feasibility team	Construction team	Operational team	Decommissioning team
Feasibility	• Real information to hand • Opportunities to inform meaningfully • Rising community expectations	- Build rapport - Multidisciplinary decision-making - Refine closure costs - Input to viability (e.g., infrastructure location, constructability) - Input to viability (resourcing, capacity, operability) - Provide direction and standards			Y		Y	
	• Short and intensive time frame • Influx of construction workers • High stress period • Local labor use • Intensive cash flow into community • Exposure to uncontrolled changes in plans	- Review process - Activity controls - Monitoring - Adhere to plans - Early advice on variations to plans that may affect closure - Engage with communities and stakeholders - Confirm construction maintains ability to close successfully - Take ownership of closure plan			Y		Y	
Construction	• Stabilization • Opportunities for continuity • Opportunities to develop long-term programs • Local labor use • Long-term capacity building • Opportunities for sustainable partnerships • Opportunity to generate familiarity and stable relationships	- Provide direction and standards - Review and advise - Review assumptions for closure - Document as-built information - Engage with community and stakeholders - Develop detailed plan - Update plan every two to three years - Implement plan - Measure against goals - Progressively improve costing accuracy - Multidisciplinary decision-making - Readiness for sudden closure		Y				
	• Local partnerships important • Risk of decline in community income • 'Last impressions'	- Engage with communities and stakeholders - Update plan - Implement plan - Implement decommissioning - Measure against goals - Provide direction and standards - Review and advise - Monitor					Y	Y
Decommissioning	• Long-term monitoring horizon • Signoff on attainment of goals • Relinquishment period	- Engage with communities and stakeholders - Measure against goals - Provide direction and standards - Review and advise - Monitor - Report		Y				Y



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Table 16.14 (a) Likelihood scales to assess risk/opportunity. (b) Description of likelihood scales. (c) Description of consequence scale (Permission: ICMM).

Likelihood Scale	Consequence Scale				
	Consequential	Limited	Overt	Significant	Extreme
1 (improbable)	Low	Low	Medium	Medium	High
2 (unlikely)	Low	Low	Medium	Medium	High
3 (possible)	Low	Medium	High	High	High
4 (likely)	Medium	Medium	High	High	Peak/Very High
5 (almost certain)	Medium	High	High	Peak/Very High	Peak/Very High

Scale	Descriptor	Description
1	Improbable	It would require a substantial change in circumstances to create an environment for this to occur, and even then this is a rare occurrence in the mining and metals industry anywhere.
2	Unlikely	There are no specific circumstances to suggest this could happen, but it has happened before at least once in the mining and metals industry.
3	Possible	There is at least a 5 per cent chance it could happen, or it has happened occasionally in other areas before, or it has occurred (albeit infrequently) in the mining and metals industry in the recorded past or risk mitigation treatment cannot reduce the inherent likelihood further.
4	Likely	There is at least a 50 per cent chance it could happen, or it has happened several times in similar areas before, or this consequence is not uncommon in the mining and metals industry or any risk mitigation treatment cannot reduce the inherent likelihood further.
5	Almost certain	Has happened/will probably happen during mine life and there is no reason to suspect it will not happen again or it has occurred in this area before.

Scale	Negative Consequence	Positive Consequence
C Consequential	Related to, in consequence of. Not inconsequential, but no more severe than that.	Related to, in consequence of. Not inconsequential but no more substantial than that.
L Limited	Some consequence, generally reversible in the short term and/or with modest application of resources (similar to daily resources operating budget for mine, if financial comparisons are appropriate).	Some consequence, not sustainable without significant ongoing application of resources.
O Overt	Consequences may be reversible, usually requiring some time and/or significant application of resources. (Similar to monthly operating budget, if financial comparisons are appropriate).	Consequences may be reversible but will generally be sustainable with only modest application of resources.
S Significant	Generally irreversible consequences, with impacts apparent for a prolonged period of time (similar time scale to mine life where time-scale comparisons are appropriate).	Generally sustainable consequences over a prolonged period of time, with little or no ongoing application of resources.
E Extreme	Irreversible consequences, impacts exceeding period similar to life of mine (where time-scale comparisons are appropriate).	Generally sustainable consequences exceeding period similar to life of mine.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 5 Knowledge platform mapping** Understanding how much is known and, equally important, how much is unknown is a crucial element in closure planning. It is important to acknowledge the validity or potential invalidity of assumptions made, because long-term closure planning should reduce the unknowns and the corresponding risk of invalid assumptions. In this simple model there are seven platforms of knowledge characterized by the information that is to hand to allow decisions to be made on whether goals are appropriate, whether they can be met, and whether the rate at which they are being met is adequate. These seven platforms are shown in table 16.15, with example numerical rankings assigned to these. By placing metrics on the levels of unknowns, the closure planner is able to track, between review periods, the robustness of information being used to refine closure plans. In addition, the closure planner is able to focus resources such as research activity on reducing unknowns in high-risk areas between review periods.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.15 Details of knowledge platform ranking (Permission: ICMM).

Characteristic	Knowledge platform	Ranking
Common Knowledge	Decisions are based on information past history, namely similar sites, data on the experience of others and company's own experience.	20
General Data	Decisions are based on site-specific baseline information, including site specific social, environmental and economic data gathered from representative areas.	30
Focused Data	Decisions are based on good-quality site-specific baseline information gathered from every specific location/aspect around the site that could be affected by moderate consequences or worse.	40
General Analysis	Decisions are based on completed studies, theoretical process or dialogue in representative areas of concern.	60
Focused analysis	Decisions are based on completed studies, theoretical processes or dialogue in every specific location/aspect around the site that could be affected by moderate consequences or worse.	70
General proof	Decisions are based on completed physical and logistical experiments on processes or trialled certain models that provide real information supporting the likelihood of success on representative areas of concern.	80
Focused proof	Decisions are based on completed physical and logistical experiments on processes or trialled certain models that provide real information supporting the likelihood of success in every specific location/aspect around the site that could be affected by moderate consequences or worse.	100

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 6 Typical headings for contextual information in a conceptual closure plan** The conceptual closure plan is primarily a vehicle for capturing and communicating the issues affecting closure. Contextual information is important in communicating these issues, as it lays out the constraints and opportunities under which the operation will develop and execute its closure plan. For practical purposes, although cross-references to key documents such as baseline studies and impact assessments should be used, a short synopsis of the key issues in the one closure planning document allows better access to these data by internal stakeholders (feasibility study managers, design teams, construction managers, etc.) as well as external ones. Select only those that apply:
 - ✓ 1. Site history
 - ✓ 2. Locality and geography
 - ✓ 3. Land tenure
 - ✓ 4. Land use
 - ✓ 5. Settlement status
 - ✓ 6. Transport networks
 - ✓ 7. Geology

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

✓ **Guideline (Toolkit) 6 Typical headings for contextual information in a conceptual closure plan**

- ✓ 8. Hydrology
- ✓ 9. Hydrogeology
- ✓ 10. Population and demographics
- ✓ 11. Household composition, density and distribution
- ✓ 12. Languages
- ✓ 13. Culture and Heritage
- ✓ 14. Community groups
- ✓ 15. Community organizations
- ✓ 16. Indigenous people
- ✓ 17. Livelihood and income streams
- ✓ 18. Industry and yield
- ✓ 19. Agriculture and yield
- ✓ 20. Per capita income
- ✓ 21. Employment rates and patterns

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 6 Typical headings for contextual information in a conceptual closure plan**
- ✓ 22. Artisanal mining
- ✓ 23. Educational facilities
- ✓ 24. Literacy and numeracy levels
- ✓ 25. Vocational skills and capacity
- ✓ 26. Health facilities
- ✓ 27. Health statistics (including HIV, malaria and tuberculosis)
- ✓ 28. Maternal health
- ✓ 29. Infant mortality
- ✓ 30. Community infrastructure
- ✓ 31. Government planning schemes
- ✓ 32. Biodiversity
- ✓ 33. Existing social legacies
- ✓ 34. Existing environmental legacies
- ✓ 35. Existing economic legacies

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 6 Typical headings for contextual information in a conceptual closure plan**
- ✓ 36. Water quality in surface and aquifer resources
- ✓ 37. Air quality
- ✓ 38. Social values requiring protection
- ✓ 39. Environmental values requiring protection
- ✓ 40. Economic values requiring enhancement
- ✓ **Guideline (Toolkit) 7 Goal setting** The worksheet Guideline (tables 16.16 and 16.17) can be used to set goals in the majority of contextual areas (see Guideline 6) and new areas. Where possible, the partial achievement of goals should be set as milestones, as the following example (shown in table 16.18) illustrates, so that there is a progressive means of assessing whether the facility is 'on track' to meet the defined closure goals.
- ✓ **Guideline (Toolkit) 8 Brainstorming support table for social goal setting (a new tool to support tool 7)** Tables 16.19(a) and 16.19(b) provide some suggested considerations for social goal setting. The list is not exhaustive, and elements should be added or deleted to suit the local conditions of the operation being considered. This format lends itself to facilitated workshops in multi-stakeholder forums and helps identify social risks and opportunities in closure planning.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.16 Worksheet that could be used to set goals in the majority of contextual areas (see Guideline 6) and also for the new areas. (Permission: ICMM).

Aspect	What must be protected?	What can be enhanced?	Goals
Land tenure			
Land use			
Settlement status			
Transport networks			
Hydrology			
Hydrogeology			
Population and demographics			
Household composition, density and distribution			
Languages			
Culture and heritage			
Community groups			
Community organizations			
Indigenous people			
Livelihood and income streams			
Industry and yield			
Agriculture and yield			
Per capita income			
Employment rates and patterns			
Artisanal mining			
Educational facilities			
Literacy and numeracy levels			
Vocational skills and capacity			
Health facilities			
Health status (including HIV, malaria and tuberculosis)			
Maternal health			
Infant mortality			
Community infrastructure			
Government planning schemes			
Biodiversity			
Existing social legacies			
Existing environmental legacies			
Water resources			
Air			
Societal values			
Environmental values			
Economic values			

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.17 Comparison of vague and well-defined closure goals (Permission: ICMM).

<i>Examples of vague closure goals</i>	<i>Examples of well-defined closure goals</i>
Land suitable for grazing	Land able to sustain grazing for nine months of the year (non-winter) for up to 100 head of cattle
Two SMEs	Two SMEs with total employment of 100 full-time equivalent local staff
Permanent healthcare facilities in village	30-bed (1% of population) permanent health care facilities with outpatient facility and maternity unit
Water supply to village	Reticulated potable water to a minimum of six standpipes in the village, with sustainable supply at 45 liters per person per day for the population of the village
Improvement in primary education	Achievement of greater than 70% attendance to Grade 4, with equivalent demographic representation of boys and girls

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.18 Setting milestones to assess progress of the defined goals (Permission: ICMM).

Closure activity	Milestone – 0%	Milestone – 25%	Milestone – 50%	Milestone goal–75%	Closure goal
Rehabilitation	Final land use plan formulated	100 hectares graded, topsoil applied and seeded	200 hectares graded, topsoil applied and seeded	300 hectares graded, topsoil applied and seeded	400 hectares graded, topsoil applied and seeded
Capacity building	Determine current and future skill base in village, including gender equality	Vocational training programs in place and full enrolment achieved	Transfer of vocational training admin. to government-owned corporation	Self-funded, self-administered vocational training institution running at full enrolment	Future skill base achieved
Tailings dam closure	Tailings impoundment location and dimensions scoped	Capping options determined and testing program commenced	Capping option chosen and material selection confirmed	Capping instituted on 50% of the (completed) cells in the tailings facility and under validation testing	Capping completed, monitoring in place



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Table 16.19(a) Suggested considerations for social goal setting (Permission: ICMM).

	<i>What social and socio-economic values or gains can be achieved</i>
Poverty	In poverty reduction?
Hunger	In hunger reduction?
Education	In education?
Gender equality	In gender equality?
Child mortality	In child mortality?
Maternal health	In maternal health?
HIV/AIDS, malaria	The management of HIV/AIDS, malaria and other diseases?
Water supply	In water supply?
Health care	In health care management?
Employment	In the employment market?
Youth employment	In youth employment?
Employability	The employability of people in the community?

Table 16.19(b) Suggested considerations for social goal setting (Permission: ICMM).

<i>Closure category</i>	<i>Typical open question</i>
	<i>What social and socio-economic values or gains can be achieved</i>
Technology	In the application of technology?
Recreation	In recreation?
Infrastructure	Through the adaptation of infrastructure?
Indigenous	Losses are inherent to indigenous affairs?
Cultural	Losses are inherent to the cultural heritage of the community?
Enterprise	Through the generation of enterprise?

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 9 Brainstorming support table for environmental goal setting (a new Guideline to support Guideline 7)** Table 16.20 provides some suggested considerations for environmental goal setting and some trigger words at the bottom of the table that can be used to formulate specific and quantifiable goals to assist in closure planning.

16.7 PLANNING FOR MINE CLOSURE

Table 16.20 Proposed considerations for environmental goal setting to formulate specific and quantifiable goals to assist in closure planning (Permission: ICMM).

<i>Closure category</i>	<i>Typical open question</i>
	<i>What environmental values, gains or losses are inherent</i>
Land resources	To land resources?
Water resources	To water resources?
Terrestrial flora	To terrestrial flora?
Terrestrial fauna	To terrestrial fauna?
Aquatic flora	To aquatic flora?
Aquatic fauna	To aquatic fauna?
Acid rock drainage	To acid rock drainage?
Air	To air?
Noise	To noise?
Waste	To waste?
Overburden dump	To the overburden dumps?
Tailings dam	To the tailings dams?

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 10 Cost risk assessment for closure** This Guideline should be read in conjunction with Financial Assurance for Mine Closure and Reclamation (ICMM, 2005). The costing of closure, particularly at the early stages of an operation's life, is aimed at adequate provisioning for closure. The contextual information in the financial assurance document should provide the practitioner with a sound understanding of why the cost of closure and the adequacy of provisioning are critical elements of closure planning. There are four structural aspects to cost estimates for each element of the closure plan:
 - ✓ I. The quantity of activity (which may be subject to variance);
 - ✓ II. The cost rate per unit quantity of the closure activity (which may be subject to variance);
 - ✓ III. An allowance that is sensible to apply (for example, using the quantities and rates for topsoil application over a four-month period leading to closure, it is sensible to expect that some rain days may cause the application of topsoil to cease, the standby time incurring a cost that is not captured in quantities and rates); and
 - ✓ IV. A contingency, which is called into play if something unplanned occurs that causes a negative consequence to be triggered, such as the unavailability of local topsoil or statutory rejection of the standard of capping.

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✓ **Guideline (Toolkit) 10 Cost risk assessment for closure**

✓ The product of quantities and rates is the base cost. An allowance is added to the base cost for conditions known from previous experience are likely to occur. A contingency is added to the base cost plus allowance to account for the possibility that something could go wrong. The worksheet (table 16.21) shows how this can be applied in the closure planning process to provide cost estimators and financial modellers with a context for conducting cost risk assessments for closure. Table 16.22 shows how a specific closure cost element (tailings dam capping) can be broken down to provide a financial modeller with information to undertake a probabilistic cost analysis.

✓ **Notes**

- ✓ 1. Wide variance here because the final tailings dam facility dimensions are unknown.
- ✓ 2. High contingency because the estimator has not yet identified where the material will be sourced from, and whether or not the appropriate material can be found on site.
- ✓ 3. As per Note 1.
- ✓ 4. Wide variance on labor costs because it has not been decided whether this will be an internal or external (contractor) cost – base rate assumes internal cost.
- ✓ 5. High contingency because it is uncertain (see Note 2) that material exists, and there is a possibility for processes such as gypsum stabilization of local or import material to meet specifications.
- ✓ 6. High variance because topsoil stockpile balance has not been carried out accurately.



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Table 16.21 The worksheet showing to guide cost estimators and financial modelers with a context for conducting cost risk assessments for closure (Permission: ICMM).

Activity	Quality			Rate		Allowance	Contingency
	Is the quantity of activity being:			Is the unit of the cost being:		What variations, from experience, should be expected?	What could go wrong in this activity to increase costs?
	Measured/ surveyed?	Estimated through calculation?	Guessed?	Extracted from quotes?	Estimated from experience?		
Goal × Activity 1							
Goal × Activity 2							
Goal × Activity 3							
Goal × Activity 4							

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.22 Illustration to understand a probabilistic cost analysis (Permission: ICMM).

Item	Quantity	– Variance in Quantity	+ Variance in Quantity	Rate	– Variance in Rate	+ Variance in Rate	Allowance	Contingency
Clay sealing material source silty clay (SC) to clay (CL) material	50,000 m ³	–5%	+30%	\$25/m ³	–0%	+10%	+10%	+50%
Clay sealing material place at 95% compaction at optimum moisture content – 1% +2% of OMC	50,000 m ³	5%	+30%	\$18/m ³	–5%	+20%	+10%	+20%
Capillary material source gravely sand d50 = 3 mm	30,000 m ³	–0%	+10%	\$6/m ³	–0%	+20%	+15%	+10%
Capillary material place, wheel loading & leveling only	30,000 m ³	–0%	+10%	\$7/m ³	–0%	+10%	+10%	+10%
Topsoil material source from stockpile	25,000 m ³	–10%	+25%	\$5/m ³	–5%	+5%	+5%	+10%

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

Table 16.23 Analysis for change in management to adopt corrective measures (Permission: ICMM).

Status change during construction	Extent of change	Could the ability to meet goals have changed [Y/N]	Should detailed closure plan be updated to capture [Y/N]	How?
Did in-migration occur?				
Were there changes in the biodiversity around the site?				
Were fauna corridors altered or transected?				
Were the courses of creeks or rivers changed?				
Did household incomes change?				
Could these changes have resulted in variability in income per household?				
Did average household sizes increase or decrease during construction?				
Were compensation payments made during construction?				
Was household relocation affected during construction?				
Could local health conditions, including the incidence of sexually transmitted diseases, have changed?				
Did local service industries benefit from, or increase as a result of, construction?				
Did local materials and supply industries benefit from, or increase as a result of, construction?				
Did employment patterns change?				
Did community organizations change, or did new community organizations form?				
Did the construction period result in new stakeholders entering the arena?				
Were there changes to the size, position or construction method applied to infrastructure during construction?				

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 11 Change management worksheet** At the juncture of the end of construction and the start of operations, many things may have changed between the pre-feasibility and operational stages. Although the impact assessment process may have been designed to scope out the majority of these changes, many actions deviate from their planned direction during the detailed design and construction periods. Design optimization, geotechnical constraints, social intervention, government statutes and other external factors can combine to ensure that one or more elements of the conceptual closure plan may have changed. Pl. refer table 16.23.

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 12 The domain model (Note: the figure is largely mining-activitybased).** A useful approach to dividing up the work to be carried out on closure is to segregate the facility into specific areas or domains. Each domain is treated as a separate detailed entity within an overall plan that deals with common issues like drainage and site monitoring.
- ✓ The following factors should be taken into account when developing a plan for each domain:
 - ✓ ● The amount and area of disturbance
 - ✓ ● Applicable legislation
 - ✓ ● Hazardous areas and risk assessments
 - ✓ ● A plan for deconstruction and decommissioning
 - ✓ ● Contamination and mitigation
 - ✓ ● End land use
 - ✓ ● Required earthworks and capping
 - ✓ ● Control of erosion
 - ✓ ● A rehabilitation plan
 - ✓ ● Monitoring
 - ✓ ● Cost estimates
 - ✓ ● Research

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 12 The domain model (Note: the figure is largely mining-activitybased).**
- ✓ Waste materials should be viewed as a resource to be used to rehabilitate other areas.
- ✓ For instance, benign overburden can be used to cap potentially acid-producing material. Proactive thinking can save the operation future liabilities. Each domain should have its own plan (see example shown in table 16.24).
- ✓ Assumptions, inclusions and exclusions should be documented. Examples of domains at a mine are:
 - ✓ • Ore processing area
 - ✓ • Shaft hoisting and headgear (underground mines)
 - ✓ • Workshops
 - ✓ • Tailings storage facility
 - ✓ • Process and raw water facilities
 - ✓ • Open voids and declines/shafts
 - ✓ • Roads (infrastructure and haul and exploration)
 - ✓ • Camps and other offices
- ✓ For accuracy, the operation should use Geographical Information System (GIS) digital terrain models and aerial photos to illustrate the domain features and boundaries; 3D models of waste dumps, voids, tailings dams and other structures are also very useful.



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Table 16.24 Illustration (an example showing how each domain should have its own plan) (Permission: ICMM).

xxx AREA							
Description							
Area of Disturbance		Void xx hectares; Waste Dumps xx hectares					
Status		Active					
Closure Date		xx void and waste rock dumps will be closed in 200x					
Infrastructure to be Retained		Final void					
OBLIGATIONS RELATING TO CLOSURE							
Subject		Obligation			Relationship to Closure		
Regulatory Condition - Rehabilitation		(F4-1) Progressive rehabilitation must commence when areas become available within the operational land.			Incorporating details and requirements for leaving voids safe and stable after closure.		
Final Land Use Objectives		Void					
EXAMPLES		(a) Safe with minimal risk to the public, native fauna and livestock					
		(b) Conceptual land use options include water bodies or partially filled water bodies. A decision will be based on the results of detailed geochemical and hydrological studies					
		Waste Dumps					
		(a) Provide an acceptable post-disturbance land use capability/suitability					
		(b) Provide acceptable, stable post-disturbance landforms					
		(c) Protect surface and groundwater quality on-site and leaving the mining lease					
		(d) Rehabilitated using technically effective and cost-efficient methods and proven engineering practices to ensure that no long-term maintenance is required beyond the post-closure phase of 5 years					
		Make the area safe with minimal risk to the public, native fauna and livestock					
CLOSURE COMPLETION CRITERIA							
Description	xxx Void	Area [ha]	14	Photo No.	Photo 1	Timing	2008-12
Engineering and Rehabilitation Activities							
(a) Excavate and haul waste rock material to construct perimeter bund walls							
(b) Fencing of void perimeters							
(c) Purchase and erect warning signs							
(d) Final pit water balance and groundwater models							
(e) Geotechnical stability assessment for long-term pit wall stability							
Statutory Sign-off							
No	Yes	Date		Document Reference No.			
✓		-		-			
Description	xx Waste Dumps	Area [ha]	28	Photo No.	Photo 1	Timing	2008-12
Engineering and Rehabilitation Activities							
(a) Selective handling of acidic rock on outer dump face of xx Dump – dispose on top surface							
(b) Re-profile xx V/D stockpile to drain towards final void							
(c) Re-profile Central dump to drain towards final void							
(d) Xx V/D stockpile – stable, minor leaching, no activities							
(e) Excavate, load and haul inert oxide waste to the reshaped xx dumps							
(f) Moderate earthworks to place store and release cover system (inert oxide rock) over PAF, reshaped dump surface and rehabilitate							
(g) Re-profile and deep rip the balance of the waste dump surface							
(h) Minor erosion control works and seeding on the balance of the waste dump surface							
(i) Geotechnical assessment to demonstrate long-term dump stability							

Table 16.24 Continued.

Statutory Sign-off	
No	Yes
✓	-
Post-Closure Activities	Void (a) Continue surface and groundwater quality monitoring for 5 years, including void water qualities and monitoring void water levels Waste Dumps Monitoring and maintenance of: (a) Revegetation works; (b) Soil erosion and soil erosion control structures; (c) Weed control in and around the rehabilitation area; and (d) Surface water quality in leachate collection ponds as per current monitoring schedules.
Specific Closure Assumptions	Void (a) The void will not be backfilled (b) The void will be allowed to flood naturally (c) Bunding and fencing will occur at closure (d) There will be no impact on groundwater as a result of water accumulating in the final void (e) The closure strategy adopted for this closure plan will be accepted by all stakeholders Waste Dumps (a) Geochemical testwork of the dumps will confirm the applicability of proposed rehabilitation methods
Closure Material Sources	Void (a) 4500 m³ of inert waste rock to construct 2 m high perimeter bund walls (b) 2500 m of fencing (c) 50 warning signs Waste Dump (a) selective handling machinery for acidic outer dump waste rock face (b) 80,000 m³ of inert waste rock from xx dump (c) Seed and fertilizer to rehabilitate 28 ha
Waste Disposal Sites	Not applicable
Other Issues	None identified
REHABILITATION COSTS (\$)	
Engineering and Rehabilitation Cost	\$500,000
Closure Administration Cost	\$20,000
Post-closure Management Cost	\$10,000
Total	\$530,000
Costs Not Included	Consultant investigations on additional geochemical testwork
Cost Saving Opportunities	(a) There may be an opportunity to generate additional cash flow by processing the I/O stockpile at the same time as reducing existing liabilities that would otherwise require re-profiling and rehabilitation. (b) Reduce existing liability by aiming to gain sign-off of the xx waste dump and final void as soon as regulators finalize the progressive rehabilitation policy.
Further Investigation/Studies Required	Long-term water quality and groundwater impacts will be needed as well as investigations on geotechnical stability of the outer dump face and verification of cover depth to restrict infiltration before regulators accept the final landforms as they are
Liabilities/Risks/Hazards	(a) The encapsulation method may not be effective in reducing acid leachate generation to acceptable levels (b) 'Hot spots' of PAF material may develop as acidic rock is exposed during any waste rock re-profiling

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.2 Guidelines/toolkit details

- ✓ **Guideline (Toolkit) 13 Biodiversity management** The ICMM Good Practice Guidance for Mining and Biodiversity was published in 2006. In fact, Biodiversity impacts play a large part in the environmental scope of mining impacts. In many remote and rural areas, biodiversity management can become one of the key elements defining successful mine closure.
- ✓ Figure 16.50 of the Good Practice Guidance provides a useful scoping Guideline (Toolkit) for capturing, for closure plan purposes, the potential impacts on biodiversity during the various stages of the facility life. Mapping of these potential impacts permits simpler access to the management, monitoring and evaluation processes that need to be included in the Conceptual Closure Plan. This figure, with site-specific impact mapping, could be included in conceptual closure plans to focus the issue.
- ✓ Figure 16.51 of the Good Practice Guidance provides more detailed information on the intersection between various mining activities and potential biodiversity impacts. This figure should be a standard part of any detailed closure plan, as it distils the closure planning issues into a practical focal point. It is particularly useful when linked to Guideline 12, the Domain Model, as this figure is largely mining activity-based.

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	MINING ACTIVITIES									
	Exploration and construction									
	Early stages of exploration									
	Access road construction									
	Land clearance for construction									
	Obtaining construction materials									
	Construction of ancillary infrastructure									
	Roads, rail & export infrastructure									
	Pipelines for slurries or concentrates									
	Energy/power & transmission lines									
	Water sources, wastewater treatment									
	Transport of hazardous materials									
POTENTIAL IMPACTS										
Impacts on terrestrial biodiversity										
Loss of ecosystems and habitats				●	●	●	●		●	●
Loss of rare and endangered species				●	●	●	●	●	●	●
Effects on sensitive or migratory species				●	●	●	●	●	●	●
Effects of induced development on biodiversity					●	●	●	●		●
Aquatic biodiversity & impacts of discharges										
Altered hydrologic regimes				●	●	●	●	●	●	●
Altered hydrogeological regimes				●		●				
Increased heavy metals, acidity or pollution				●		●	●	●	●	●
Increased turbidity (suspended solids)				●	●	●	●	●	●	●
Risk of groundwater contamination				●		●	●	●	●	●
Air quality related impacts on biodiversity										
Increased ambient particulates (TSP)				●	●	●	●	●	●	●
Increased ambient sulfur dioxide (SO ₂)							●		●	●
Increased ambient oxides of nitrogen (NO _x)							●		●	●
Increased ambient heavy metals									●	
Social interfaces with biodiversity										
Loss of access to fisheries					●	●		●	●	●
Loss of access to fruit trees, medicinal plants					●	●	●	●	●	●
Loss of access to forage crops or grazing					●	●	●	●	●	●
Restricted access to biodiversity resources					●	●	●	●	●	●
Increased hunting pressures				●	●	●	●	●	●	●
Induced development impacts on biodiversity					●	●	●	●	●	●

Figure 16.50 A useful scoping Guideline for capturing, for closure plan purposes, the potential impacts on biodiversity during the various stages of the facility life (Permission: ICM).

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	MINING ACTIVITIES									
	Ore processing and plant site	Plant site, materials handling, etc.	Extraction and waste rock storage	Rock blasting and ore removal	Mine dewatering	Placer and dredge mining	Ore stockpiling	Pyrometallurgical processing	Hydrometallurgical processing	Use and storage of process chemicals
POTENTIAL IMPACTS										
Impacts on terrestrial biodiversity										
Loss of ecosystems and habitats	●	●			●					●
Loss of rare and endangered species	●	●			●					●
Effects on sensitive or migratory species	●	●	●				●	●		●
Effects of induced development on biodiversity										
Aquatic biodiversity & impacts of discharges										
Altered hydrologic regimes	●	●	●	●	●			●		●
Altered hydrogeological regimes		●	●	●	●			●		●
Increased heavy metals, acidity or pollution	●	●	●	●	●	●		●	●	●
Increased turbidity (suspended solids)	●	●	●	●	●	●		●		●
Risk of groundwater contamination	●	●	●	●		●		●	●	●
Air quality related impacts on biodiversity										
Increased ambient particulates (TSP)	●	●	●			●	●			●
Increased ambient sulfur dioxide (SO ₂)							●			
Increased ambient oxides of nitrogen (NO _x)							●			
Increased ambient heavy metals	●	●	●			●	●			●
Social interfaces with biodiversity										
Loss of access to fisheries	●	●								●
Loss of access to fruit trees, medicinal plants	●	●								●
Loss of access to forage crops or grazing		●		●		●		●		●
Restricted access to biodiversity resources		●								●
Increased hunting pressures	●									
Induced development impacts on biodiversity										

Figure 16.51 Detailed information on the intersection between various mining activities and potential biodiversity impacts (Permission: ICMM).

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.3 Glossary

- ✓ **Care and maintenance** – The period following temporary cessation of operations when infrastructure remains largely intact and the site continues to be managed.
- ✓ **Closure Planning** – A process that extends over the mine life cycle and that typically culminates in tenement relinquishment. It includes decommissioning and rehabilitation. The term closure alone is sometimes used to indicate the point at which operations cease, infrastructure is removed and management of the site is largely limited to monitoring.
- ✓ **Decommissioning** – The process that begins near or at the cessation of mineral production and ends with the removal of all unwanted infrastructure and services.
- ✓ **Deterministic estimates** – Estimates of value (cost or benefit) of the outcome of an event occurring, expressed as a single mean or mode value and a range of single values (e.g., minimum, maximum).

16.7.3 The Integrated Mine Closure Planning Guidelines (Toolkit)

16.7.3.3 Glossary

- ✓ **Probabilistic estimates** – Estimates of value (cost or benefit) that account for the likelihood of occurrence and the range of values of the outcomes. Values are expressed through a statistical analysis (e.g., Monte Carlo simulation) using a statistical distribution over the range of possible values accounting for the probability and timing of the event occurring.
- ✓ **Reclamation/Rehabilitation** – Terms used interchangeably to mean the return of disturbed land to a stable and productive condition.
- ✓ **Relinquishment** – Formal approval by the relevant regulating authority indicating that the completion criteria for the mine have been met to the satisfaction of the authority.
- ✓ **Stakeholder** – A person, group or organization with the potential to affect or be affected by the process or outcome of mine closure.
- ✓ **Internal stakeholders** - It includes various groups or teams within the mining company such as: exploration team, corporate team, feasibility team, construction closure team, operational team, decommissioning/closure team.

Thank You

Queries??

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